

28/3/2026

Machine UNlearning

Security: The adversarial data generated is similar to the original data. The model will make incorrect predictions.

Detect $\rightarrow$ Delete $\rightarrow$ Machine Unlearning

Privacy: Delete the user data to avoid risk of data leakage.

Usability: Trying to buy a phone $\rightarrow$ Bought a phone

Accuracy: Forget this biased datasets $\rightarrow$ fairness/
Accuracy

Machine unlearning

1. Training of the ML models is stochastic in nature.
2. Training is an incremental process
3. Catastrophic unlearning

Notations:

Z space of examples / data items

Z^* set of all possible training datasets

particular dataset $\underline{D} \in Z^*$

hypothesis space $\underline{H}$

function $\underline{A} : Z^* \rightarrow H$

Machine Unlearning

unlearning mechanism function $\underline{U}$

unlearning set $\underline{D_f} \in D$

$\downarrow$
unlearn

Stochastic Algorithm

unlearning model $\underline{U}(D_f, D, A(D)) \in H$

$\Pr(A(D))$ distribution of all models obtained using learning algorithm A on dataset D .

$\Pr(D_f, D, A(D))$ distribution of forgotten models on dataset D

$\Pr(D_f, D, A(D))$ distribution of machine

Unlearning model

Def. 1: Given any learning algorithm A , dataset D and an unlearning set $D_f \in D$, $V(\cdot)$ is an exact unlearning process if

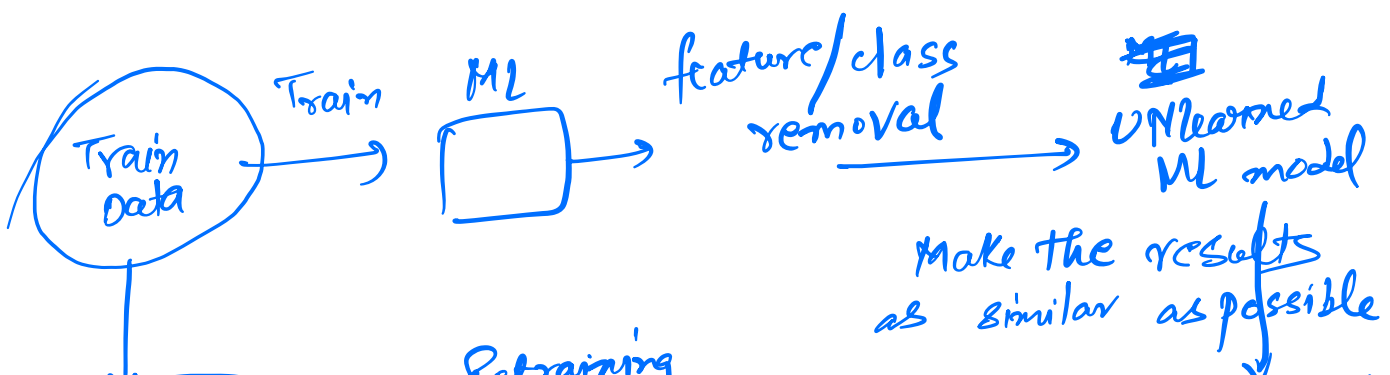
$$\Pr(A(D \setminus D_f)) = \Pr(V(D, D_f, A(D)))$$

Def. 2: Given a learning algorithm A , V is an exact unlearning process if
 $\forall T \in H, D \in Z^*, D_f \subset D$

$$\Pr(A(D \setminus D_f) \in T) = \Pr(V(D, D_f, A(D)) \in T)$$

Approximate unlearning: Def 3 / 4

Framework of Machine Unlearning:



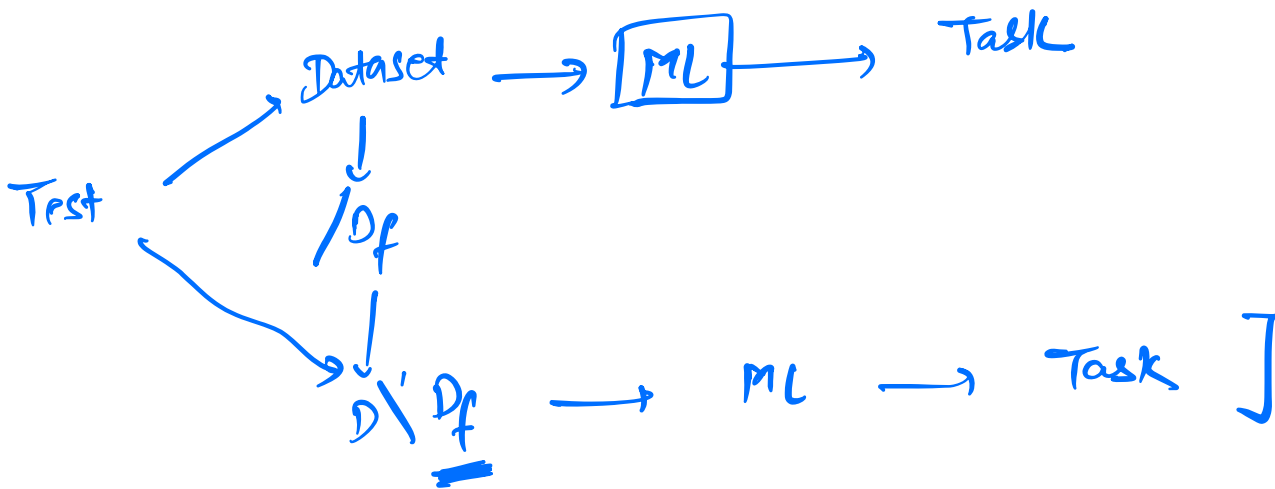


Requests for Machine Unlearning:

1. Feature Deletion: Certain samples needs to be removed from dataset.
2. Class Deletion: The data belongs to one or more classes.

Design Requirements:

1. Verifiability:



2. Consistency



3. Accuracy: Comparable to the retrained model.

Validation of Machine Unlearning:

1. Membership inference attack

Detect $\rightarrow$ Target model leaking data

$\rightarrow$ Shadow models

$\downarrow$
Trained on a new
set of data items.

Predictions

$\rightarrow$ Attack model

Trained on this

Tested

Data

$D \times P_f$

2. Back door attacks: It is the injection of a backdoor into data to spoof a machine learning model.

... make correct preds for

Spoof a model → make clean data
incorrect preds. for toxic data

Backdoor trigger:

